

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

School of Computing — Department of Computer Science & Engineering

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Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit II | Session 2 | SLO 1: Data Types, Variables & Scope (Code: 2021)

Q: Activity: Implement a Java program demonstrating primitive data types, variable scope, dynamic initialization, and the distinction between local, instance, and static variables.

Ans: Java Implementation:

```
public class VariableScopeDemo {
    // Static (class) variable - single copy per class
    static int classCounter = 0;

    // Instance variable - unique per object
    String objectLabel;

    public VariableScopeDemo(String label) {
        this.objectLabel = label;
        classCounter++;
    }

    public void compute() {
        // Local variable - exists only in method scope
        byte byteVal = 100;
        double dynamicVal = Math.sqrt(byteVal);
        System.out.println("Label: " + objectLabel);
        System.out.println("Dynamic Local Value: " + dynamicVal);
        System.out.println("Class Counter (Static): " + classCounter);
    }

    public static void main(String[] args) {
        VariableScopeDemo d1 = new VariableScopeDemo("Object Alpha");
        d1.compute();
        VariableScopeDemo d2 = new VariableScopeDemo("Object Beta");
        d2.compute();
    }
}
```